Control Tower — Enterprise User Manual

PM Control Tower (Control Tower) · Platform version 3.1.0 · October 2026

Audience: executive sponsors, PMO leaders, administrators, project managers and delivery teams.

Status: authoritative consolidation. Cross-references the numbered document suite (01–13) and specialist guides (AI-BYOK, FLOW import, EMAIL-REMINDERS, ANALYTICS, BUSINESS-PLAN).

Platform	Control Tower — Enterprise Project, Program & Portfolio Management
Brand lockup	Optinus Solutions × Czarnikow (header); login shows "Control Tower"
Runtime	Node ≥ 20.9 (Next.js standalone, Bun optional), Socket.IO realtime gateway
Data	SQLite (default) or PostgreSQL (Vercel/hosted deployments), Prisma schema
Scale in repo	119 data models · 186 API route files · 209 view modules · 28 computation engines · 43 permissions · 8 role templates · 48 sidebar tabs · 100 AI features · 143 automated tests

How to read this manual. Sections 1–4 tell you *what you get and how the work flows*. Section 5 is the architecture with diagrams. Sections 6–7 are the business case and cost. Section 8 is setup, end to end. Statements marked **[E]** are estimates/assumptions, **[R]** are planning ranges, **[V]** are verified vendor/market figures (sourced in the business plan).

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1. Benefits — what the platform delivers

One platform, one truth. Control Tower replaces the fragmented stack (spreadsheet plans + task tool + BI extracts + slide-deck assembly) with a single system where the plan, the actuals, the money and the reporting come from the same database — so the number on the executive slide is the same number in the task list.

1.1 Benefits by stakeholder

Persona	Primary benefit	Evidence in the product
Executive / Sponsor	Real-time portfolio truth without waiting for a monthly pack; faster, better-grounded decisions	Executive Control Tower, Leadership Control Tower (20 tabs in 4 groups), Investment & Business Case (NPV/IRR/payback), EVM, alerting
PMO leader	Standardised way of working across all projects; automated status production; defensible governance	Reporting Center (library, builder, board packs, scheduled runs, subscriptions), Stage Gates, Governance, Data Integrity, 70 PMO templates
Project manager	One workspace from charter to closure — scope, schedule, cost, risk, change, status — with AI drafting the first 80%	Project Register → WBS → Schedule (CPM/Gantt/baselines) → EVM/Health/RAID/Changes, AI PM Assistant, 100-feature AI Feature Studio
Delivery team	Clear personal work surface; no tool-hopping	Boards (Kanban/Calendar/Workload), Tasks, Agile Delivery (sprints), Work Inbox, Focus Planner, Timesheets
Finance controller	Actuals that tie to the plan; forecast discipline; billing control	Financials, EVM (CPI/SPI/EAC/ETC/VAC), Clients & Billing, Expenses, Investment cases
Field / site teams	Site reality flows into the same project record	Field Operations (RFIs, daily logs, punch lists), Proofing & Approvals
Compliance / Auditor	Immutable trail, permission boundaries, evidence on demand	Audit log, RBAC (43 permissions), Integrity (freshness, evidence bundles, doctor checks)
IT / Security	Self-hosted control, no vendor lock-in, BYOK AI, standard stack	Docker/PM2/Bun/Windows-pack deployments, SQLite/Postgres, JWT + TOTP + SCIM + SSO (MSAL), encrypted AI key vault

1.2 Platform-level benefits

1. **Single source of truth** — 119 models on one core; every tab reads and writes the same records. No reconciliation between a scheduling tool, a task tool and a BI extract.
2. **Effort replacement, not just tool replacement** — status production, pack assembly and roll-ups are computed (engines) and rendered (Reporting Center, PPT export of 42 views) instead of being hand-built. The business case models this as the dominant cost line (PMO reporting effort: status-quo ~$480k/3yr at 100 users vs ~$60k with Control Tower) **[E]**.
3. **Deterministic calculation, AI for narrative** — CPM, EVM, Monte Carlo, PERT, finance and leveling are computed in code; the AI layer only narrates numbers the client computed and is forbidden (system prompt, enforced) from inventing metrics.
4. **Bring-your-own-key AI at zero licence cost** — one provider key (Gemini default, OpenAI/Groq swappable) powers the assistant and all 100 features; no per-seat AI add-on, no subscription inside the platform, key encrypted server-side and never exposed to the browser.

5. **Governance by construction** — RBAC on every API (43 permissions, 8 role templates), audit events on mutations, rate limiting, stage gates, change control and a data-integrity layer (freshness, evidence, doctor) make answers defensible in an audit.
6. **Deployment freedom** — Docker Compose, bare metal (PM2/nginx), Bun, a one-click Windows pack, or Vercel + Postgres. Data stays where you put it.
7. **Automation of the routine** — Control Automations, webhooks, integrations and the daily consolidated e-mail reminder (Microsoft Graph) keep work moving without a human chasing.

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2. Feature catalogue — every capability

2.1 Portfolio & project core (Plan)

- **Project Register** — portfolio inventory with codes, health, owners, dates, budget.
- **Requirements** — requirement backlog with status and traceability to delivery.
- **WBS** — hierarchical work breakdown (drag-and-drop tree) that feeds schedule and cost.
- **Tasks** — cross-project task list with effort, assignees, due dates, status, priority.
- **Agile Delivery** — epics, sprints (one active per project), story points, backlog grooming, sprint board with drag-and-drop and server-recomputed committed points.
- **Schedule** — CPM engine (dependencies, constraints, lag), Gantt with today marker, critical path, progress fills and milestone diamonds; **Milestones** and **Baselines** snapshots for variance; **Planning Lab** for what-if/estimate tooling (PERT, Monte Carlo).
- **Portfolios / Programs / Projects** — three-level hierarchy with roll-up of health, budget and schedule across levels.

2.2 Execution (Execute)

- **Boards (omni views)** — the same tasks as Kanban (status lanes), Calendar (month grid, weekends shaded) and Workload (resource × week heatmap, 4–26 weeks, over-allocation red).
- **Resources** — resource register (avatars seeded from employee code), assignments, capacity vs demand.
- **Timesheets** — own time entry with draft/submit workflow, approval queue, actuals feeding cost and EVM.
- **Work Inbox** — personal triage of assigned actions, approvals and alerts.
- **Focus Planner** — personal timeboxing of the work that matters this week.

2.3 Control & finance (Control)

- **Financials** — budgets, costs, commitments per project/program.
- **Clients & Billing** — clients, invoices (server-computed totals, paid-invoice lock), expenses.
- **Investment & Business Case** — cases with NPV, IRR, payback, sensitivity (finance engine).
- **EVM** — PV/EV/AC, SPI/CPI, EAC/ETC/VAC with configurable alert thresholds.
- **Health** — composite health scoring and trend.
- **RAID** — risks, issues, assumptions, decisions with owners, severity, mitigation tracking.
- **Change Control** — change requests with impact and decision trail.
- **Stage Gates** — gate definitions, evidence, go/no-go decisions per phase.
- **Governance** — governance controls and compliance view for the PMO framework.
- **Data Integrity** — freshness signals, evidence bundles, doctor checks (orphans, deadline breaches, unassigned effort) so numbers can be trusted.
- **Prioritization** — weighted scoring / optimizer to rank competing investments.
- **CTO Tower** — technology/estate oversight view.
- **Field Operations** — RFIs (auto-numbered, aging), daily logs, punch lists, BIM iframe.

2.4 Strategy & knowledge

- **Feedback & Roadmap** — feedback posts with votes, triage into roadmap lanes.
- **Docs & Knowledge** — spaces, pages, markdown editor (institutional memory).
- **Message Boards** — threaded discussions, replies, pins (decision record).
- **Proofing & Approvals** — review rounds with annotations and approval decisions.

2.5 Intelligence

- **Executive Control Tower** — live KPIs, RAG donut, investment view, top risks, alerts, capacity, EVM bars, hours trend, governance queue.
- **Leadership Control Tower** — 20-tab executive/PMO pack in 4 groups.
- **Reports** — report suite across the portfolio.
- **Reporting Center** — sections: Library · My Reports · Report Builder · Board Packs · Scheduled · Subscriptions · History (template library documented at 177 report templates).
- **Analytics** and **PMO Analytics (Insights)** — trends, distributions, insights reports.
- **AI PM Assistant** — conversational drafting grounded in live register data.
- **AI Feature Studio** — 100 ranked features (f001–f100) across 10 categories, each a one-button draft generator with client-computed arithmetic (EVM, PERT, EAC/ETC/VAC, phasing, cash-flow, velocity, ratios) and an honest fallback when the AI key/quota/network is missing.
- **Export** — universal PowerPoint export: any of 42 views as a native 16:9 deck.

2.6 Connect

- **Integration Hub** — connector catalog to external systems.
- **Control Automations** — event/rule automations (e.g. overdue escalation).
- **Notifications** — in-app notification centre and preferences.
- **Webhooks** — outbound HTTP hooks on platform events.
- **AI Connector Gateway** — central AI provider routing/status.
- **Extension Hub** — optional capability extensions.

2.7 Administration

- **Users** (activate/deactivate, role change, activity), **Roles** (8 templates + custom, 43-permission matrix), **Templates** (70 PMO project templates, apply-to-new-project), **Audit** (server-side filtered event log with before/after JSON), **Settings** (env readiness, AI provider BYOK card, probes, entity counts), **Fields** (custom-field dictionary), **Tabs** (tab builder), **SCIM/SSO** identity integration.

2.8 Platform engines (`src/lib/engines/` — 28)

`cpm` (critical path) · `evm` · `montecarlo` · `leveling` (resource) · `calendars` (working-day index) · `finance` (NPV/IRR/payback) · `prioritization` · `scenario` · `rollup` · `executive-portfolio` · `leadership` · `steering` · `health` · `governance` · `benefits` · `doctor` (data health) · `freshness` · `evidence` · `identity` (TOTP/SCIM) · `metadata` (custom fields) · `custom-fields` · `automations` · `connectors` · `timesheet` · `calibration` · `pptx` + `pptx-views` (export) · `zip` .

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3. Tab linkage — every tab to its outcome

The sidebar has **8 groups and 48 tabs**. Each row links: **tab → route → what you do → the immediate outcome/artifact → the permission required**. Section 4.3 shows how the outcomes chain end to end.

3.1 Executive

Tab	Route	What you do there	Outcome / artifact	Permission
Executive Control Tower	`#/dashboard`	Read live KPIs, RAG, investment, risks, alerts, capacity, EVM, governance queue	Instant portfolio situational awareness; alert-driven interventions	`executive.view`

3.2 Plan

Tab	Route	What you do there	Outcome / artifact	Permission
Project Register	`#/register`	Create/curate projects, codes, owners, budgets, dates	The portfolio inventory every other tab keys from	`project.view`
Requirements	`#/requirements`	Capture and status requirements	Scope baseline with traceability	`project.view`
WBS	`#/wbs`	Build the work breakdown tree	Scope structure feeding schedule & cost	`project.view`
Tasks	`#/tasks`	Manage work items, owners, effort, dates	Execution backlog and actuals source	`project.view`
Agile Delivery	`#/agile`	Run epics/sprints/backlog, point story cards	Sprint execution and velocity history	`project.view`
Schedule	`#/schedule`	Build CPM schedule, Gantt, critical path	Committed dates, critical path, slippage signal	`project.view`
Milestones	`#/milestones`	Plan key dates	Milestone plan feeding gates and exec view	`project.view`
Baselines	`#/baselines`	Freeze a schedule snapshot	Variance reference (plan vs actual)	`project.view`
Planning Lab	`#/tools`	Run what-if / estimate tools (PERT, Monte Carlo)	Estimate confidence ranges for commitments	`project.view`

3.3 Portfolio

Tab	Route	What you do there	Outcome / artifact	Permission
Portfolios	`#/portfolios`	Group projects into portfolios, watch roll-ups	Investment container + aggregate health	`portfolio.view`
Programs	`#/programs`	Manage program-level structure and roll-up	Programme oversight and dependency view	`program.view`
Projects	`#/projects`	Open the full project workspace (tabs per project)	The project's single operating picture	`project.view`

3.4 Execute

Tab	Route	What you do there	Outcome / artifact	Permission
Boards	`#/boards`	Move work across Kanban / Calendar / Workload views	Flow control, over-allocation spotted early	`project.view`
Resources	`#/resources`	Maintain the resource register and assignments	Capacity vs demand picture (avatars, codes)	`resource.view`
Timesheets	`#/timesheets`	Log and approve time	Actuals that drive cost and EVM	`timesheet.own` (+ `timesheet.approve`)
Work Inbox	`#/inbox`	Triage assigned actions and approvals	Zero-miss personal throughput	`inbox.use`
Focus Planner	`#/planner`	Timebox the week	A realistic personal delivery plan	`inbox.use`

3.5 Control

Tab	Route	What you do there	Outcome / artifact	Permission
Financials	`#/financials`	Track budget/cost/commitments	Cost baseline and spend visibility	`financial.view`
Clients & Billing	`#/clients`	Manage clients, invoices, expenses	Revenue control and billing evidence	`financial.view`
Field Operations	`#/field`	RFIs, daily logs, punch lists, BIM	Site-to-office data loop	`project.view`
Investment & Business Case	`#/investment`	Build cases with NPV/IRR/payback	Fund-or-kill evidence	`financial.view`
EVM	`#/evm`	Read SPI/CPI/EAC/ETC/VAC	Objective performance measurement	`evm.view`
Health	`#/health`	Review composite health/trend	Early-warning indicator set	`evm.view`
RAID	`#/raid`	Maintain risks, issues, assumptions, decisions	Exposure register with mitigations	`project.view`
Change Control	`#/changes`	Raise and decide change requests	Scope/decision audit trail	`project.view`
Stage Gates	`#/gates`	Run gate reviews with evidence	Go/no-go decisions per phase	`project.view`
Governance	`#/governance`	Apply governance controls and review compliance	Compliance posture for the PMO framework	`project.view`
Data Integrity	`#/integrity`	Freshness, evidence bundles, doctor checks	Trust layer: "can I defend this number?"	`integrity.view`

Tab	Route	What you do there	Outcome / artifact	Permission
CTO Tower	`#/technology`	Technology/estate oversight	Technology risk and investment view	`integration.view`
Prioritization	`#/prioritization`	Score and optimise competing initiatives	Ranked, defensible investment order	`portfolio.view`

3.6 Strategy & Knowledge

Tab	Route	What you do there	Outcome / artifact	Permission
Feedback & Roadmap	`#/feedback`	Capture/vote feedback, triage to roadmap	Demand-driven product/delivery direction	`inbox.use`
Docs & Knowledge	`#/docs`	Author and organise knowledge pages	Institutional memory	`docs.use`
Message Boards	`#/boards-hub`	Discuss and pin threads	Durable decision record	`boards.use`
Proofing & Approvals	`#/proofing`	Review rounds with annotations, decide	Sign-off evidence on deliverables	`project.view`

3.7 Intelligence

Tab	Route	What you do there	Outcome / artifact	Permission
Leadership Control Tower	`#/reports/leadership`	Read the 20-tab, 4-group leadership pack	Board-ready leadership reporting	`reports.view`
Reports	`#/reports`	Run the report suite	Standard analytical outputs	`reports.view`
Reporting Center	`#/reporting`	Library, builder, board packs, schedules, subscriptions	Recurring, self-service reporting (no analyst queue)	`reports.view`
Analytics	`#/analytics`	Explore trends and distributions	Insight charts for decisions	`reports.view`
PMO Analytics	`#/insights`	PMO insights hub and insights reports	Programme/portfolio diagnostics	`reports.view`
AI PM Assistant	`#/assistant`	Ask for drafts grounded in live data	First-draft status/briefs in seconds	`ai.use`
AI Feature Studio	`#/ai-features`	Trigger any of 100 ranked AI features	Domain drafts (status, risk, cost, agile…) with computed facts	`ai.use`

3.8 Connect

Tab	Route	What you do there	Outcome / artifact	Permission
Integration Hub	`#/integrations`	Configure connectors	Systems kept in sync	`integration.view`
Control Automations	`#/automations`	Define event rules (e.g. overdue escalation)	Work that happens without chasing	`integration.view`

Tab	Route	What you do there	Outcome / artifact	Permission
Notifications	`#/notifications`	Preferences and notification history	Awareness without inbox noise	`inbox.use`
Webhooks	`#/webhooks`	Outbound event endpoints	Downstream system triggers	`integration.view`
AI Connector Gateway	`#/ai-gateway`	AI provider routing/status	Central AI infrastructure control	`integration.view`
Extension Hub	`#/extensions`	Enable optional extensions	Capability added without a release	`integration.view`

3.9 Administration (header → Settings / admin routes)

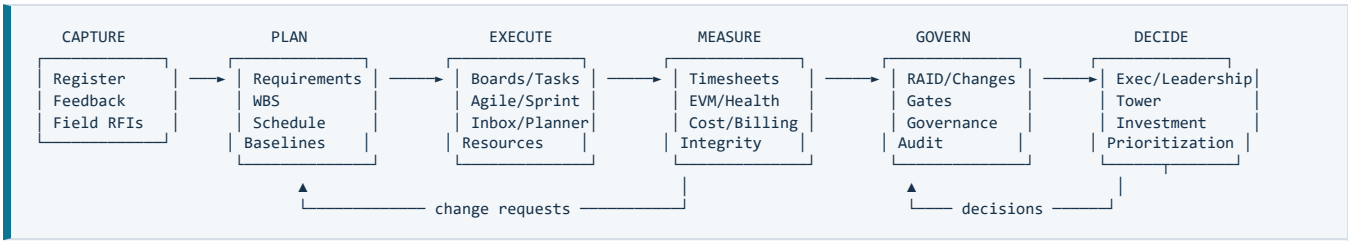
Area	Route	Outcome	Permission
Admin Center	`#/admin-center`	Administration landing	`admin.users` etc.
Users	`#/admin/users`	People on/offboarded, roles assigned	`admin.users`
Roles	`#/admin/roles`	Permission matrix realised per role	`admin.users`
Templates	`#/admin/templates`	New projects born standardised (70 templates)	`admin.templates`
Audit	`#/admin/audit`	Who changed what, when, with before/after	`admin.audit`
Settings	`#/admin/settings`	Env readiness, AI key (BYOK), probes, branding	`admin.config`
Custom fields	`#/admin/fields`	Metadata dictionary (schema without code changes)	`admin.config`
Tab builder	`#/admin/tabs`	Workspace composition per need	`admin.config`

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4. End-to-end working design

4.1 The operating loop

Control Tower encodes one management loop, and every tab belongs to a stage of it:



Data lifecycle: a project created in the Register fans out to WBS → Schedule (CPM) → Baselines; execution writes Tasks/Sprints/Timesheets; the engines compute EVM, health, cost and roll-ups; the integrity layer marks freshness and evidence; Reporting Center and the AI layer render it; decisions flow back as changes, gate outcomes and reprioritisation.

4.2 Personas in the loop (day-in-the-life)

1. **Team member (morning)** → Work Inbox shows 3 assigned actions → Focus Planner timeboxes 2 → moves a Kanban card → logs 6.5h in Timesheets (draft → submit).
2. **PM (daily)** → Register/Workspace update → Schedule recalculates critical path → RAID updated → Board reflects flow → Automations flag an overdue milestone → AI Feature Studio drafts the weekly status (f001) from computed facts → sends via Reporting Center schedule.
3. **PMO (weekly)** → Leadership pack refreshed → Data Integrity doctor run → governance queue cleared → Reporting Center board pack exported (PPT) → subscriptions delivered by e-mail.
4. **Executive (monthly)** → Executive + Leadership towers → drill to Investment/EVM → Prioritization re-ranks → gate decisions recorded → audit trail complete.

4.3 How the tabs chain (linkage end to end)

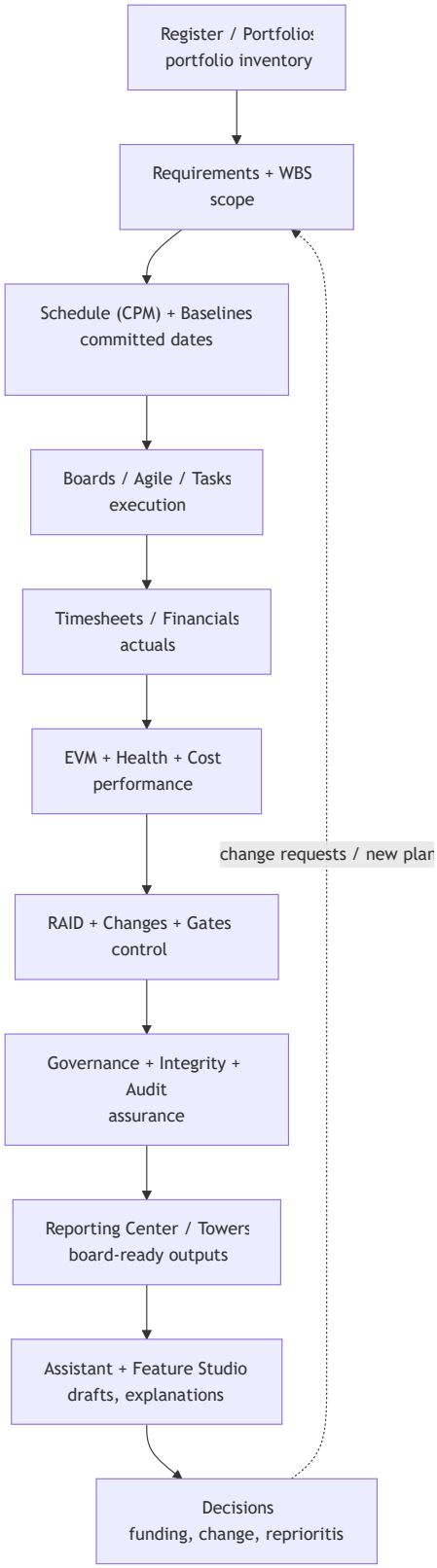


Figure 1 — How the tabs chain (linkage end to end)

4.4 System working design (runtime behaviour)

1. **Authentication** — login issues a JWT held in an httpOnly cookie; the same secret material authenticates the realtime gateway (`REALTIME_SECRET_KEY = JWT_SECRET`). Optional TOTP MFA, SSO via MSAL (OIDC/Azure AD), SCIM provisioning.

2. **Routing** — hash router (`#/segment/sub`); `registry.tsx` maps segment → view component; the shell composes sidebar + topbar + command palette (⌘K). Role-aware home route redirects users without `executive.view` to the first workspace they can use.
3. **API pipeline** — every `/api/**` route runs: session/JWT → RBAC permission check → zod validation → business logic (engines/services) → audit event → rate limit → realtime emit → typed JSON response.
4. **Engines** — deterministic computation (CPM, EVM, Monte Carlo, finance, roll-up, leveling…) runs server-side in TypeScript; tests assert the maths (143-test suite).
5. **Realtime** — API publishes domain events (`project:changed` , `audit:created` , …) to the Socket.IO gateway (`:3003`) via an internal emit call; the browser subscribes to rooms and updates views + toast notifications without reload.
6. **AI** — client calls `askAIResult` (a "use server" action): session → `ai.use` RBAC → rate-limit (10/min) → decrypt BYOK key from the encrypted vault → provider adapter (Gemini/OpenAI/Groq, 30s timeout) → ledger entry in `AIExecution` . It never throws: failures return a typed status and an honest fallback string shown in the UI.
7. **Persistence** — Prisma over SQLite (absolute `DATABASE_URL`) or PostgreSQL on hosted deployments; additive `db push` schema sync (no migration history), reference data via seeds.

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5. Architecture and diagrams

5.1 System context

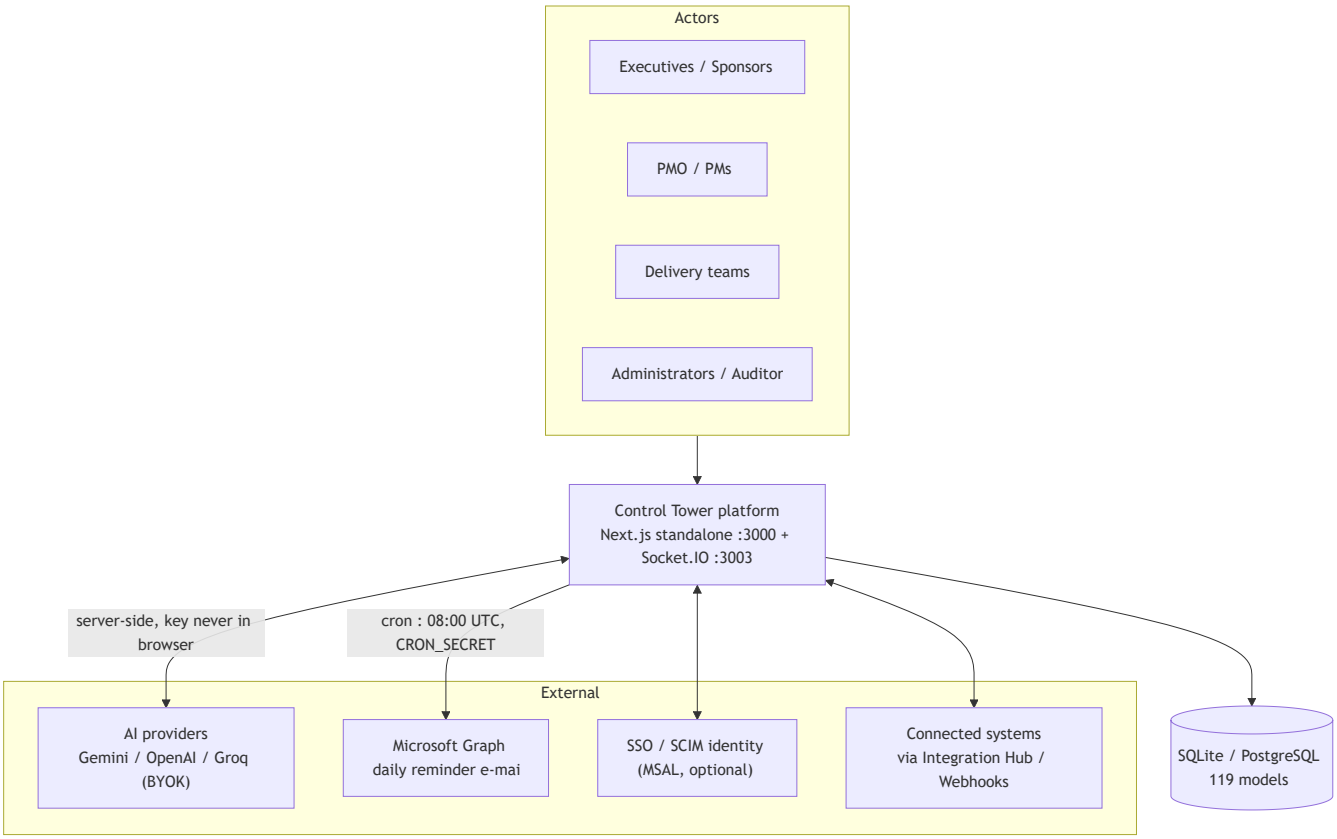


Figure 2 — System context

5.2 Layered architecture

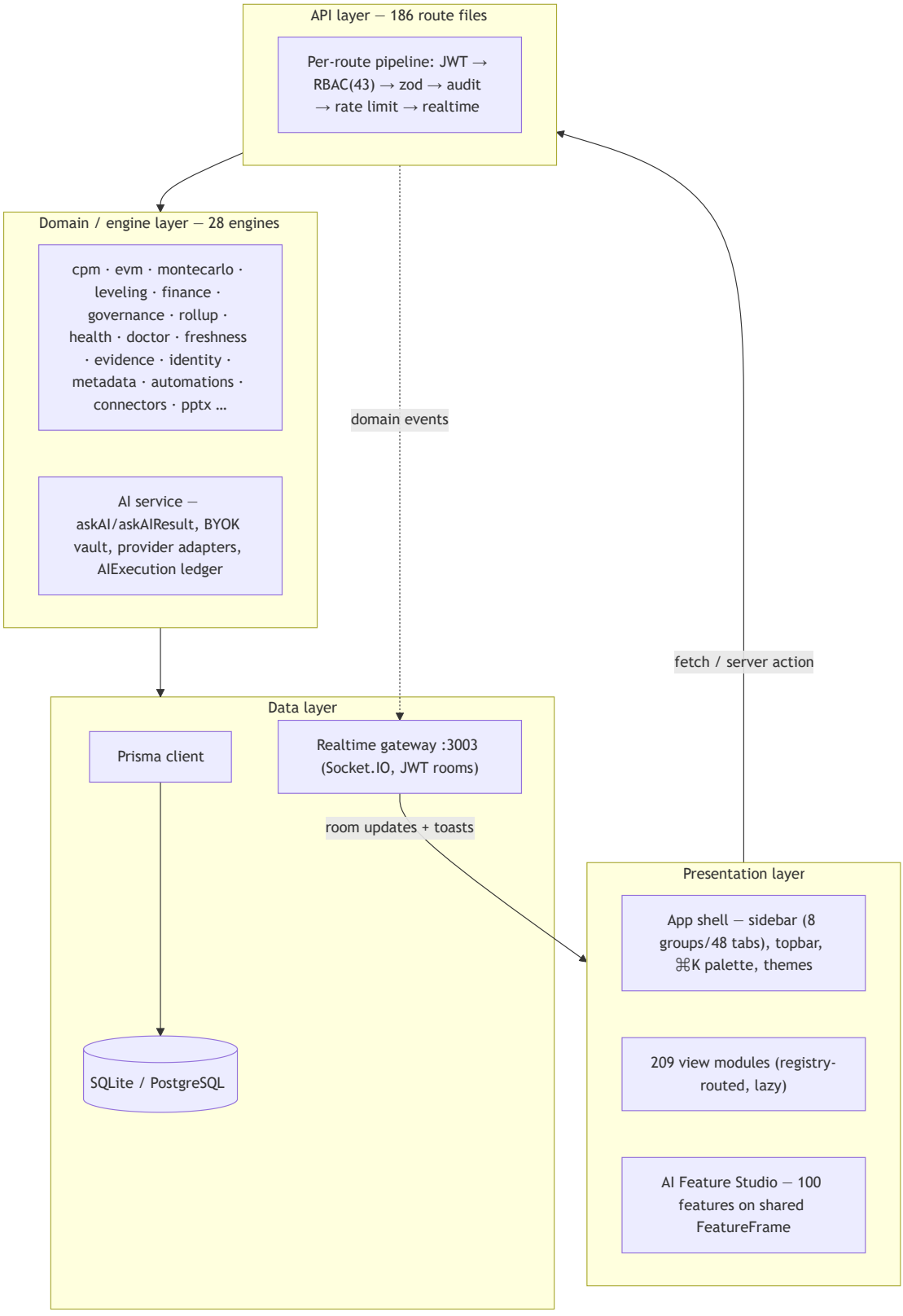


Figure 3 — Layered architecture

5.3 Deployment topology

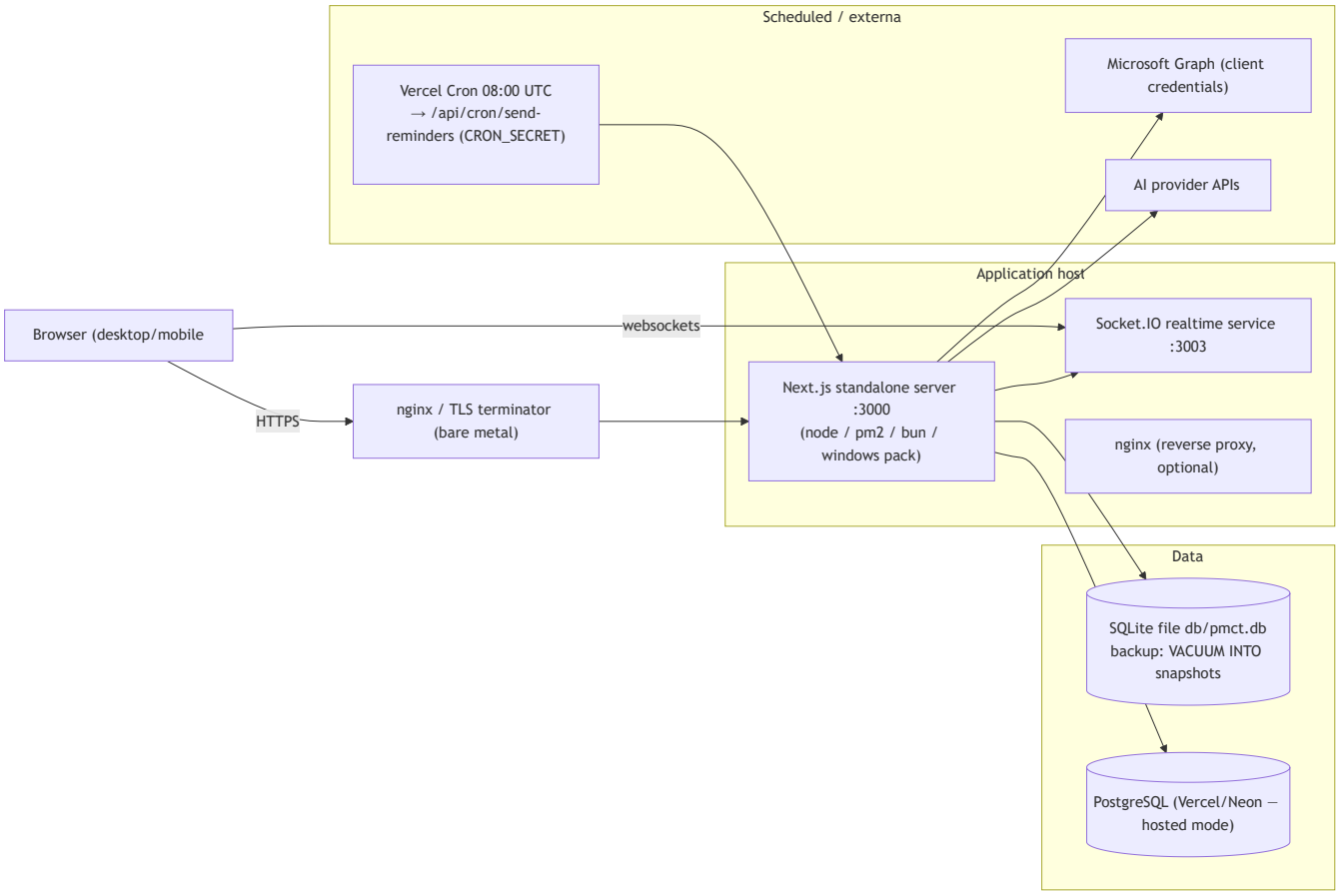


Figure 4 — Deployment topology

Deployment modes (details: `docs/01-DEPLOYMENT-GUIDE.md`): A Docker Compose (recommended) · B bare metal (Node 20 + PM2 + nginx) · C Bun runtime · D Windows deployment pack (one-click exe) · E Vercel + Postgres.

5.4 Request pipeline (sequence)

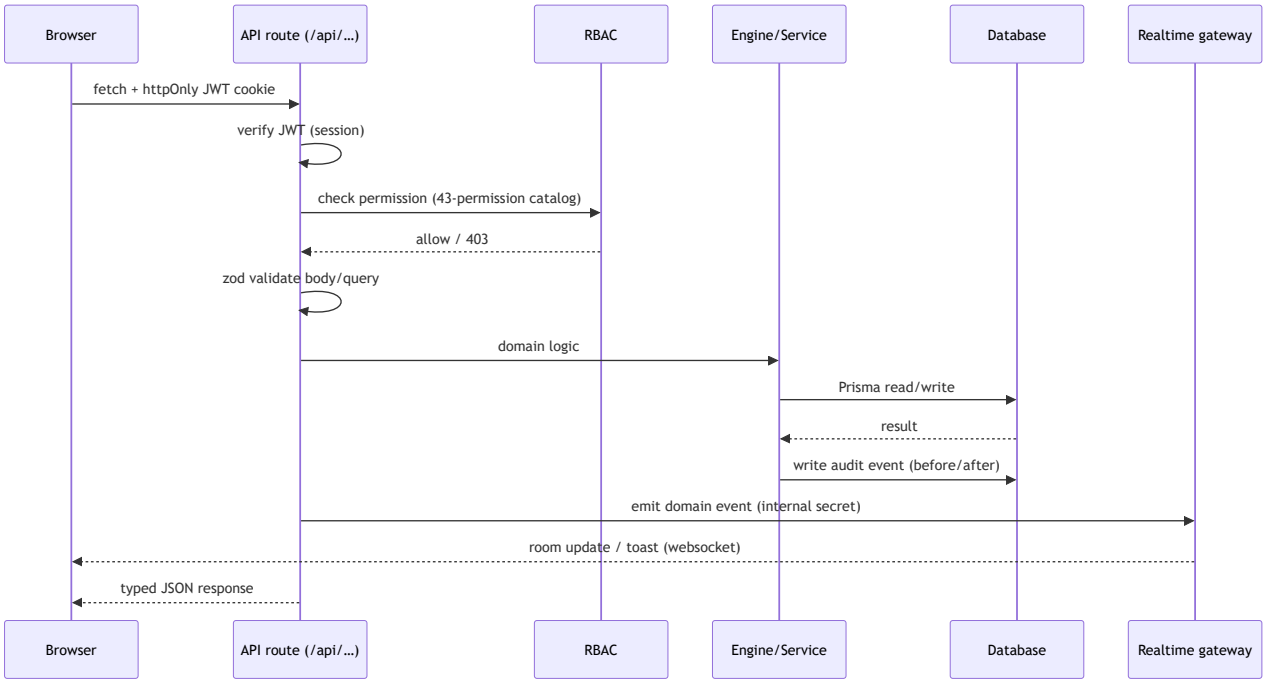


Figure 5 — Request pipeline (sequence)

5.5 Authentication & authorisation

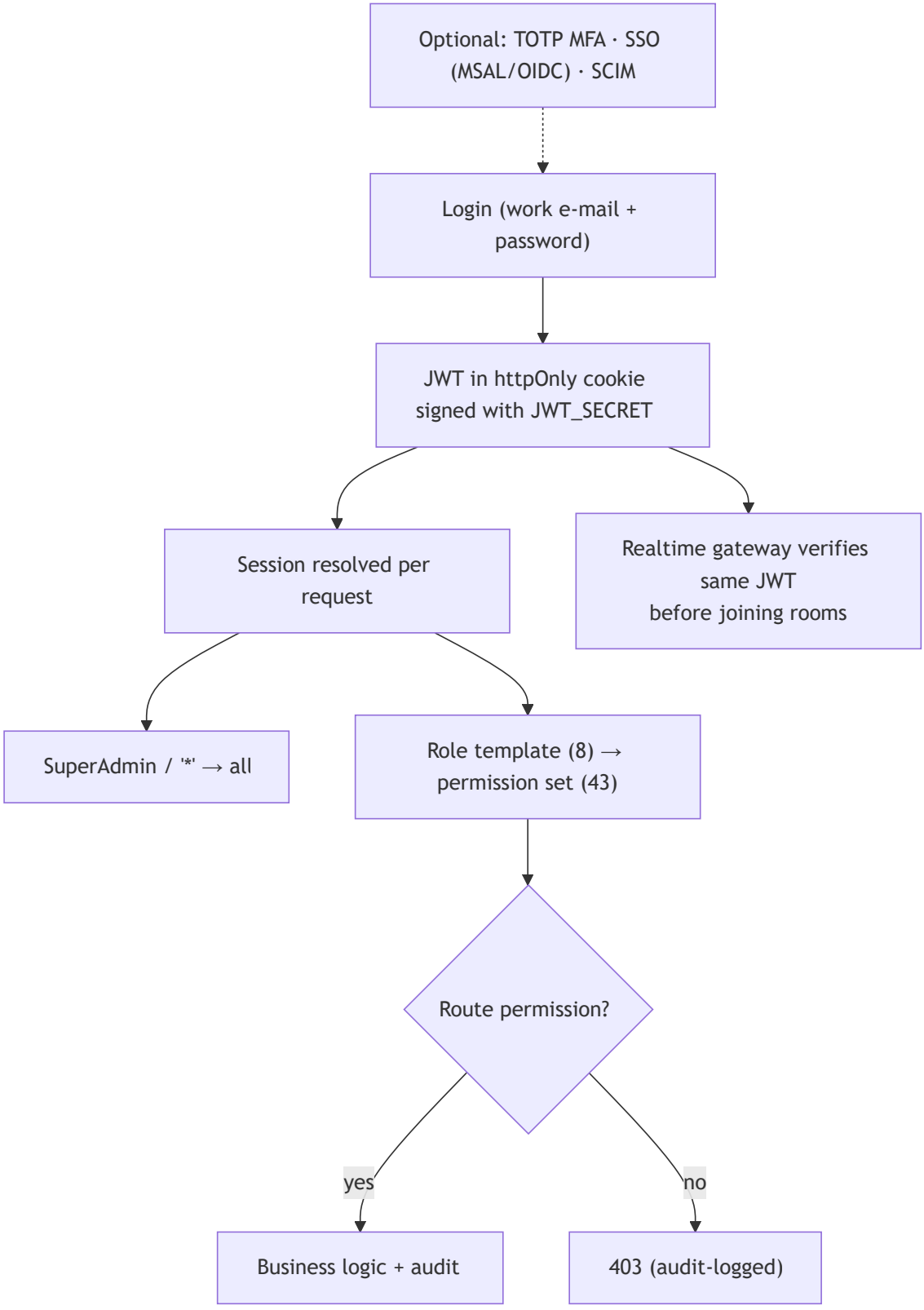


Figure 6 — Authentication & authorisation

5.6 Realtime flow

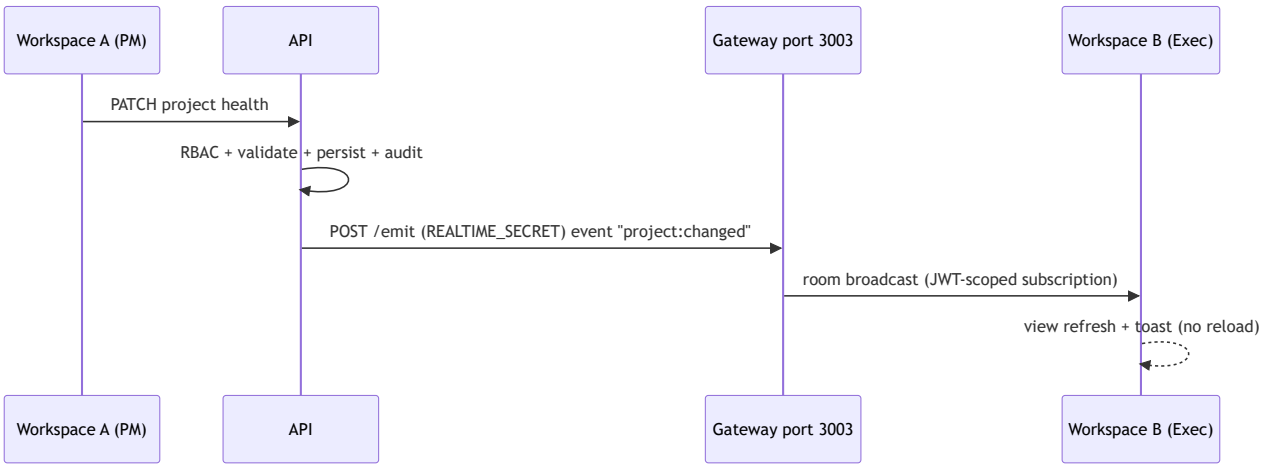


Figure 7 — Realtime flow

5.7 AI (BYOK) flow

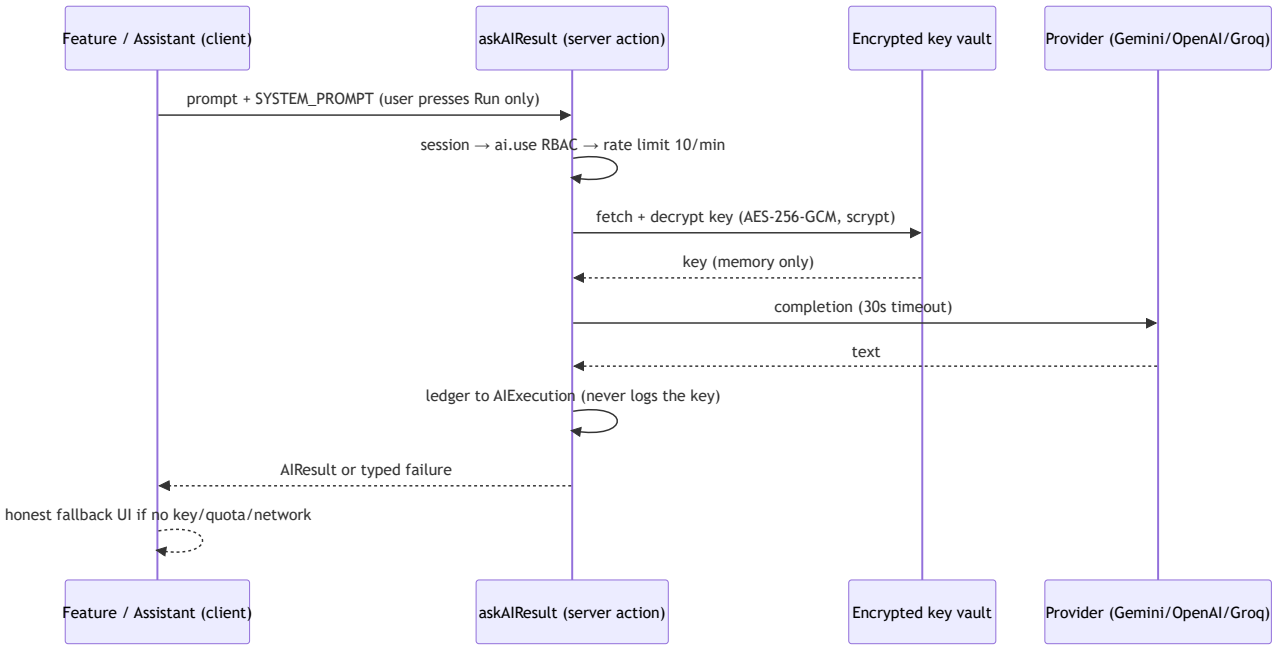


Figure 8 — AI (BYOK) flow

Rules enforced by design: no subscription/seat cost inside the platform; key never reaches localStorage or any browser code; `askAIResult` never throws; client-side code computes all arithmetic (EVM, PERT, EAC/ETC/VAC, phasing, cash-flow, velocity, ratios) and the model narrates computed facts.

5.8 FLOW planning-workbook import

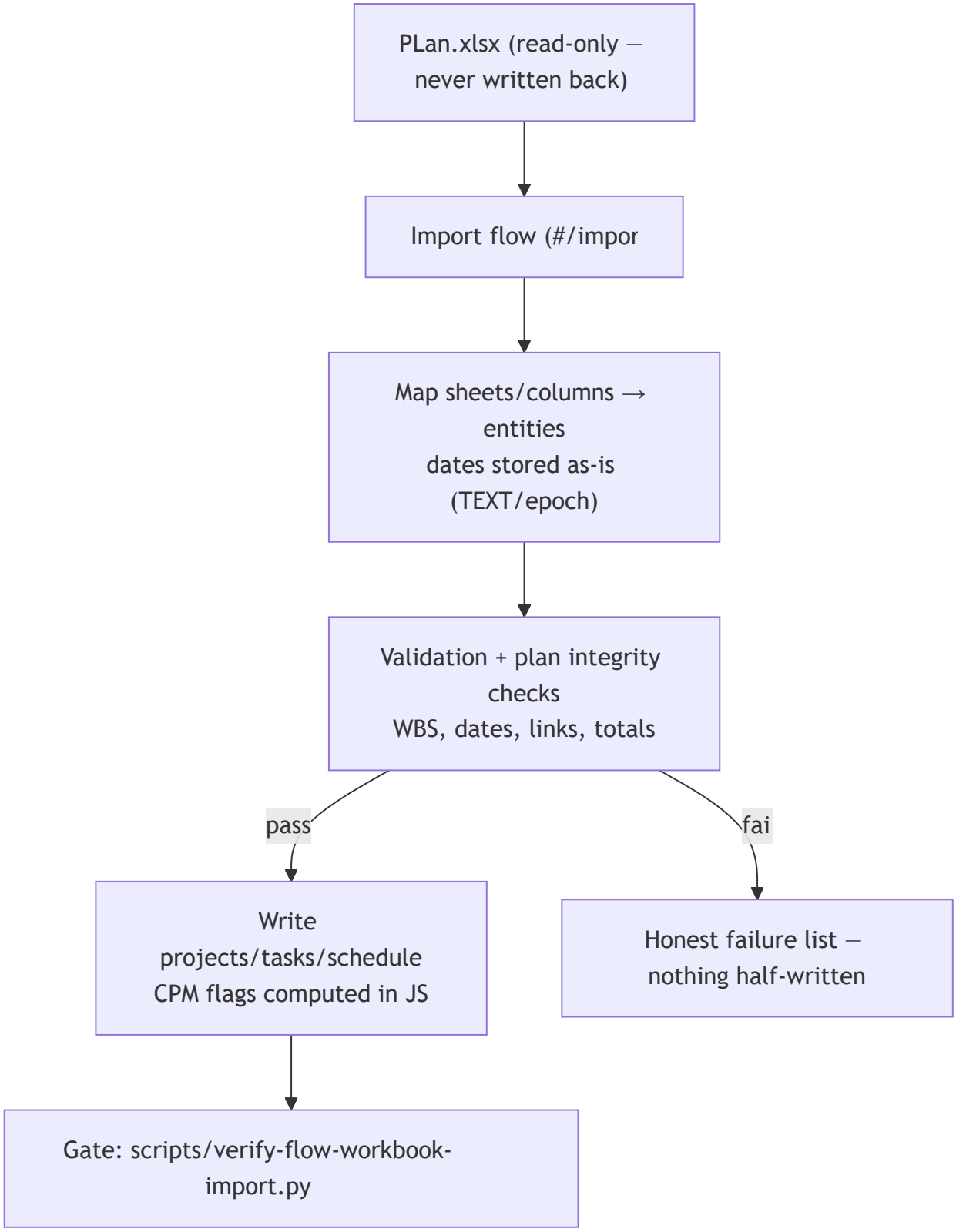


Figure 9 — FLOW planning-workbook import

Storage rule: imported workbook dates stay in their stored form (TEXT) while Prisma dates are epoch milliseconds — dates are never reformatted on write; window filtering happens in JS.

5.9 Daily reminder e-mail (cron)

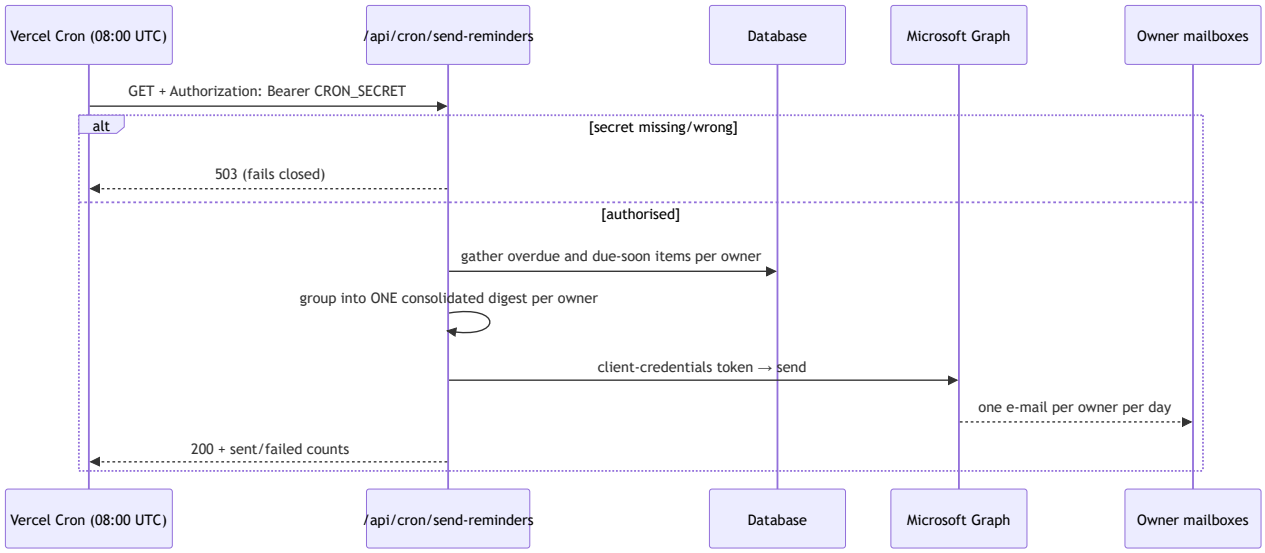


Figure 10 — Daily reminder e-mail (cron)

`dryRun=1` returns the would-be e-mail without sending. Unconfigured Graph credentials deliberately return 503 (visible, not silent). Setup: `docs/EMAIL-REMINDERS-GUIDE.md`.

5.10 Data model overview (119 models)

Domain	Representative models
Structure	Portfolio, Program, Project, WBS node, Requirement, Milestone, Baseline
Work	Task, Sprint, Epic, Story points fields, Board lane config, Inbox item, Planner slot
Time/cost	Timesheet/entry, Budget/cost lines, Invoice(+lines), Expense, Activity
Control	Risk, Issue, Assumption, Decision, Change request, Gate, Governance record, Evidence
Quality	Integrity/freshness record, Doctor finding, Audit event, Proof round/annotation
Knowledge	Doc page, Message thread/reply, Feedback post/vote, Roadmap item
Field	RFI, Daily log, Punch item
Platform	User, Role, Permission assignments, Custom field, Template, Webhook, Automation, Integration, AI connector config, AIExecution ledger, Notification

5.11 Repository layout & build

```
src/
  app/            Next.js app router — page.tsx (auth gate), layout, api/**/route.ts (186)
  components/pmct/ shell (sidebar/topbar/palette), kit (design system), brand
  views/          domain view modules (admin, plan, portfolio, execute, control,
                  intelligence, connect, knowledge, strategy, ai/features (100)) — 209 files
  lib/            engines/ (28), ai/ (service, providers, vault), rbac, auth, audit, realtime client
prisma/           schema.prisma (119 models)
mini-services/    realtime/index.ts (Socket.IO gateway)
scripts/          seed*, sync-rbac, backup-app.ps1, verify-flow-workbook-import.py
tests/engines/    143 deterministic tests (no DB required)
docs/             this manual + numbered suite 01–13 + specialist guides
build             next build → .next/standalone (self-contained server + public/)
```

6. Business case

6.1 The problem

1. **Fragmentation** — plans in one tool, tasks in another, money in spreadsheets, reporting in slides. Every hand-off is a re-keying opportunity and a reconciliation cost.
2. **Reporting glue is the biggest cost** — status production and pack assembly consume PMO capacity that should be spent steering. The business plan models **~2 FTE of reporting effort (~$480k over 3 years at 100 users)** in the status-quo stack **[E]**.
3. **Slow, unsafe decisions** — by the time a pack is assembled the data is stale; without RBAC/audit/evidence, answers cannot be defended in a review or audit.
4. **Tool sprawl cost** — the same 100-user organisation spends $10k–$65k/yr on tooling alone (Jira Premium $20k, Asana $36k, MS Project P3 $41k, Aha! $81k — Year-1 TCO tables) before counting people **[V/E]**.
5. **AI is either unavailable or metered per seat** — competitors charge $9–$28/user AI add-ons **[V]**, so the teams who need drafting help the most ration it.

6.2 Solution mapping

Problem	Control Tower capability	Measurable outcome
Fragmented tools	One platform, one database (119 models), 48 tabs on one core	Reconciliation work → ~0; one login
Reporting glue cost	Reporting Center (library/builder/board packs/scheduled/subscriptions), Leadership pack, PPT export (42 views), AI drafts	PMO reporting effort 0.25 FTE vs 2 FTE **[E]**
Stale/undefendable data	Real-time towers, Integrity layer (freshness/evidence/doctor), audit + RBAC	Decisions on current data; evidence bundles on demand
Licence sprawl	Replaces PPM + agile + docs + BI glue + field ops	See §7.5 benchmark
Metered AI	BYOK AI: assistant + 100 features at token cost only	AI adoption without procurement cycles
Adoption failure	Role-aware home, per-user inbox/planner, honest empty states, WCAG 2.2 AA	Contribution layer feeds the executive layer

6.3 Value model (illustrative calculator) [E]

```
Annual benefit  = B1 + B2 + B3 + B4
  B1 Tool consolidation    = Σ(current tool subscriptions displaced)
  B2 Reporting effort      = (FTE_before − FTE_after) × loaded cost
  B3 Faster decisions      = value of earlier intervention
                             (fewer late gate surprises; estimate 1–2% of portfolio spend)
  B4 Avoided rework        = fewer data-driven errors (integrity layer + single source)
Annual cost    = licence/hosting (§7) + implementation + admin time + AI tokens
ROI            = (Annual benefit − Annual cost) / Annual cost
Payback        = Implementation cost / Monthly benefit
```

Worked example — 100 users, 3-year horizon (business plan Part 12) [E]:

	Status-quo stack	Control Tower
Subscription (3 yr)	~$101k (Jira + Smartsheet + Power BI)	$75.6k (Business tier)
Implementation/integration/training	~$90k	~$52k
PMO reporting effort	**~$480k (2 FTE)**	**~$60k (0.25 FTE)**
Data reconciliation fixes	~$60k	~$5k
3-year TCO	**~$730k**	**~$193k**

→ **~$537k avoided over 3 years (~$179k/yr)** at 100 users, before faster-decision value. Against Tier-2 PPM (Planview ~$620k, ServiceNow ~$860k for the same span) the gap widens. Sensitivity: if reporting-effort saving is only half real, payback still occurs inside year 1 on tool substitution alone.

6.4 Risks and mitigations

Risk	Mitigation
Adoption (another tool to learn)	Role-aware home, inbox-first workflow, 70 PMO templates, training pack (`docs/training/`), phased rollout (§8.11)
Data migration quality	FLOW workbook importer with validation + plan-integrity gate; honest failure lists
Single-vendor dependency	Self-hosted code + export (PPT, API, webhooks); no proprietary lock-in
AI accuracy	Deterministic engines compute; AI narrates; never-invent-metrics system prompt; human-in-the-loop "draft then review"
Security	RBAC/audit/rate limits/TOTP/SCIM/SSO; BYOK key vault; fails-closed cron; secrets via env only
Key-person dependency	Documented suite (01–13), runbooks, 143 automated tests as regression net

6.5 Success measures

KPI	Baseline (typical)	Target with Control Tower
Hours per monthly status pack	40–80 h (manual)	< 8 h (scheduled + AI draft) **[E]**
PMO hours/week on reporting glue	~2 FTE	≤ 0.25 FTE **[E]**
Projects with a current baseline	40–60%	≥ 90%
Overdue items without owner	common	0 (doctor + automations)
Tool subscriptions in the PMO stack	4–6	1 + BYOK AI
Audit evidence retrieval	days	minutes (evidence bundles)

---

7. Cost

> All figures are internal planning figures unless tagged. **[V]** verified vendor pricing as captured in `docs/BUSINESS-PLAN-2026-2027.md` Part 10 (Oct 2026), **[E]** estimate, **[R]** planning range.

7.1 What running Control Tower costs

Cost line	Self-hosted (typical)	Hosted (Vercel + Postgres)	Notes
Platform licence	**$0 in your own deployment** (code is yours) — or commercial tiers if procured (§7.2)	same	No per-seat metering inside the product
Infrastructure	$6–$24/mo VPS, or $0 marginal on existing Windows/server hardware **[E]**	Vercel Pro ~$20/mo + Postgres $0–$50/mo **[E]**	SQLite needs no DB server
Operations	~2–4 h/month admin **[E]**	~1–2 h/month **[E]**	backups are scripted (one command)
AI tokens (BYOK)	$20–$150/mo at ~5k runs/month **[E]**	same	Your own provider key; metered per call
E-mail reminders	$0 (Microsoft Graph inside your M365 tenant)	same	client-credentials app registration
Implementation	one-time: days, not quarters (§8)	same	business plan models $30k at 100 users for full program **[E]**

AI token maths [E]: each AI run ≈ 3–4k input tokens (system prompt + context) + 0.5–1k output. 5,000 runs/month ≈ 17–25M tokens/month. At mid-tier rates ($0.10–$1.50/M input, $0.40–$6/M output) that is roughly **$20–$150/month** — choose model/price in Administration → Settings → AI provider. Every call is ledgered (`AIExecution`) so spend is observable, and rate limits (10/min/user) cap runaway usage.

7.2 Commercial pricing (if procured as a product)

From the business plan (Part 14) — annual prepaid per user:

	Starter	Professional	Business	Enterprise	Enterprise Plus
Annual	**$9**/user/mo (≤25)	**$15** (≤100)	**$21** (≤500)	**$29** (≤2,000)	custom
Platform fee	—	—	—	$75k+/yr unlimited users	$150k+/yr
AI	included	included	included	included	included
SSO	—	✓	✓	✓	✓ (+SCIM/MFA enforce)
Self-hosted	—	—	✓	✓	✓ (+air-gapped)
Support	community	business hours	8–5	24–5, SLA 99.5%	24–7, SLA 99.9%, CSM

Perpetual/one-time licence from ~$75k (Part 15). AI is deliberately included (competitors charge $9–$28/user add-ons) and SSO is included from Professional (competitors gate it to Enterprise) — two repricing anchors versus the market.

7.3 Benchmark — same organisation, same year **[V/E, business plan Part 11]**

Year-1 TCO (subscription + AI + implementation), 100 users:

Tool	Y1 TCO	vs Control Tower
Jira Premium	$20,065	1.4×
Asana Advanced	$35,986	2.5×
Monday Pro / Smartsheet	$27,360	1.9×
MS Planner + Project P3	$41,420	2.9×
Aha! Premium	$81,420	5.7×
Planview Portfolios **[R]**	~$126,000	8.8×
ServiceNow SPM **[R]**	~$240,000	16.7×
Primavera P6 EPPM **[R]**	~$262,500	18.3×
Control Tower (Business)	**$48,000** (incl. $30k implementation)	1×

At 25 users: PMCT $12,000 vs Asana $8,996/Wrike $9,000/Planview ~$31,500 — and PMCT includes PPM depth (CPM/EVM/gates) those work-management tools lack. At 500+ users the platform-licence model (from $75k/yr unlimited) diverges sharply from per-seat scaling (Asana ~$180k/yr at 500 before enterprise tiers) **[E]**.

7.4 Three-year TCO (100 users) **[E, business plan Part 12]**

	Status quo stack	ServiceNow SPM	Planview	
3-year TCO	~$730k	~$860k	~$620k	**~$193k**
Dominant line	PMO reporting effort $480k	subscription $360k	subscription $216k	subscription $75.6k

7.5 What you get at every price point

All tiers ship the core loop: register → WBS → CPM schedule → baselines → tasks/boards → timesheets → EVM/health/RAID/changes → governance/audit → reporting → AI. Tiers add scale (projects/portfolios), depth (prioritization optimizer, investment cases, integrity layer, 177-template reporting, automations/webhooks, connector count), and enterprise controls (SSO/SCIM/self-hosted/support SLA).

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8. Setup — installation to go-live

8.1 Prerequisites

- Node.js **≥ 20.9** (or Bun, or the Windows pack which bundles everything)
- 4 GB RAM / 2 vCPU is ample for a departmental deployment **[E]**
- A host reachable by your users (VPS, existing Windows server, or Vercel)
- Optional: Microsoft 365 tenant (for reminder e-mails), an AI provider account (for BYOK), TLS terminator (nginx/Caddy) or Vercel HTTPS
- Read `docs/.env.example` — the template for `.env`

8.2 Choose a deployment mode

Mode	Best for	Key steps (detail: `docs/01-DEPLOYMENT-GUIDE.md`)
A. Docker Compose (recommended)	Servers, VMs	`cp .env.example .env` → set secrets → `docker compose up -d` → build + realtime + seeded DB
B. Bare metal (PM2 + nginx)	Existing Linux estates	`npm ci` → `.env` → `npm run db:generate && npm run db:push` → `npm run db:seed` → `npm run build` → `pm2 start npm --name pmct -- start` → proxy :3000 with TLS
C. Bun runtime	Bun shops	`bun install` → seed → `npm run build` → `npm run start:bun` (+ `npm run realtime`)
D. Windows pack	Air-gapped/on-prem one-click	unzip `dist/PM-Control-Tower-Deploy-*.zip` → `start.cmd` → health-check script; autostart included
E. Vercel + Postgres	Serverless hosting	attach Postgres (Neon), set `JWT_SECRET`, push schema (`vercel.json` build does `db push` + seed-if-empty), deploy

8.3 Environment variables (complete)

Variable	Required	Purpose
`DATABASE_URL`	yes	`file:/absolute/path/to/db/pmct.db` (absolute in production) or `postgresql://` when hosted
`JWT_SECRET`	yes	32+ random bytes (`openssl rand -hex 32`) — signs sessions
`REALTIME_SECRET_KEY`	yes	must equal `JWT_SECRET` — gateway verifies the same JWTs
`REALTIME_SECRET`	yes	shared secret for API → gateway `POST /emit`
`REALTIME_URL` / `REALTIME_PORT`	yes	internal gateway address (default `http://127.0.0.1:3003`)
`PORT` / `HOSTNAME`	defaults	standalone server bind (3000 / 127.0.0.1)
`WEB_ORIGIN` / `PMCT_CONFIG_JSON`	optional	public origin + config bootstrap (templates)
`AZURE_TENANT_ID`, `AZURE_CLIENT_ID`, `AZURE_CLIENT_SECRET`	for e-mail	Graph client-credentials app
`MAIL_SENDER_EMAIL`	for e-mail	shared mailbox that sends the digest

Variable	Required	Purpose
`CRON_SECRET`	for e-mail	bearer token Vercel Cron sends; route fails closed (503) without it
`EVM_SPI_THRESHOLD` / `EVM_CPI_THRESHOLD`	optional	alert thresholds (default 0.9)

Never commit `.env`. Secrets are read server-side only (the AI key is additionally encrypted AES-256-GCM at rest and never returned to the browser).

8.4 Initialise data

```
npm ci
npm run db:generate          # prisma client
npm run db:push              # schema sync (additive)
npm run db:seed              # reference data + leadership seed
npm run seed:command-center  # demo/operational dataset (clients, invoices, agile, field…)
npm run sync:rbac            # upsert 43 permissions into 8 roles (no reseed)
npm run seed:integrity       # integrity/evidence demo data (optional)
npm run seed:strategy        # strategy/roadmap data (optional)
npm run build && npm start    # production on :3000   (+ npm run realtime for :3003)
```

8.5 First-run checklist

1. **Sign in** with the seeded admin (quick-account shortcuts exist on the login page for demo tenants) and change credentials immediately on real deployments.
2. **Roles & users** — invite users (`#/admin/users`), assign one of the 8 role templates (Executive, PMO Administrator, Portfolio Manager, Program Manager, Project Manager, Finance Controller, Team Member, Auditor) or a custom role.
3. **Settings** — verify env readiness (✓/! indicators), probes (realtime, database), branding.
4. **Import your plan** — FLOW planning workbook import (never writes back to `PLan.xlsx`), then run `python scripts/verify-flow-workbook-import.py` → must print `FLOW_WORKBOOK_VERIFICATION=PASS`.
5. **AI key** — Administration → Settings → *AI provider (Bring Your Own Key)*: provider (Gemini default / OpenAI / Groq), model, write-only key, **Test**. Status shows masked `••••1234` only. Key is encrypted at rest; audit-logged masked.
6. **E-mail reminders** — register the Azure AD app, fill Graph vars + `MAIL_SENDER_EMAIL` + `CRON_SECRET`, schedule cron at 08:00 UTC; test with `dryRun=1`.
7. **Backup** — `powershell -File scripts\backup-app.ps1` creates two date-stamped archives (`pmct-backup-*.zip` = DB snapshot + .env + schema; `pmct-app-backup-*.zip` = full source). Schedule it (Task Scheduler / CI) — restore drill documented in the operations runbook.
8. **Smoke test** — log in as each role: sidebar shows only permitted tabs; run one report; move one board card; draft one AI feature (expect the honest "not configured" state until step 5); check zero console errors.

8.6 Verification gates (quality bar for every change)

```
npx tsc --noEmit            # type safety (0 errors)
npx eslint .                # lint (0 problems)
npm test                    # 143 deterministic tests (engines, RBAC, AI, calendar, CPM…)
python scripts/verify-flow-workbook-import.py   # plan integrity (prints PASS)
```

8.7 Rollout plan (indicative) [E]

Phase	Weeks	Activity	Exit criteria
1 — Foundation	1–2	Deploy, secrets, roles, admin settings, backup	All roles log in; probes green
2 — Data	2–3	Import plan, resources, financials; integrity doctor clean	Doctor findings = 0 critical
3 — Pilot	3–5	1–2 teams: boards, timesheets, RAID, first scheduled report	Pilot status pack auto-generated
4 — PMO scale-out	5–8	Gates, governance, Reporting Center schedules, AI key, PPT export	Monthly pack ≤ 8 h
5 — Org-wide	8–12	Remaining users, automations, webhooks, integrations	Adoption KPIs (§6.5) trending

9. Security, governance and operations

Security posture (summary — detail: `docs/10-SECURITY-GUIDE.md`) - Authentication: JWT httpOnly cookies, argon/bcrypt-class password storage, optional TOTP MFA, SSO via MSAL (OIDC), SCIM provisioning. - Authorisation: 43 permissions × 8 role templates; super-admin wildcard; enforced **server-side on every route** (client hiding is cosmetic, not the control). - Audit: every mutation writes an audit event with before/after JSON; audit UI has server-side filters, pagination, evidence export. - Abuse controls: per-route rate limits (auth, AI 10/min, exports), zod validation, size limits. - Secrets: env-only; AI key additionally encrypted (AES-256-GCM + per-record scrypt salt/IV), masked in UI/logs; cron fails closed without `CRON_SECRET`. - Transport: TLS via nginx/Caddy or Vercel; realtime authenticates with the same JWT material.

Governance. Stage gates, change control, the governance controls view, RAID discipline and the integrity layer (freshness, evidence bundles, doctor) together give the PMO a defensible control framework; the audit trail covers who-decided-what-when.

Operations (detail: `docs/09-OPERATIONS-RUNBOOK.md`) - Health: `/api/system/ready`, Settings probes, realtime LIVE indicator. - Backups: dual date-stamped archives (system + full app); SQLite snapshots via `VACUUM INTO` are safe while the app runs; restore drill = unpack system zip → point `DATABASE_URL`. - Upgrades: take backup → `git/npm` update → `db:push` (additive) → re-seed if empty → restart; keep `PMCT_DIST_DIR` isolation if a build runs beside a live server. - Monitoring: process manager (PM2/systemd) restarts, log files, audit/AIExecution tables for activity volume, cron response codes (200 vs 503).

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10. Glossary and document index

Glossary. *BYOK* — bring-your-own-key (your AI provider key, encrypted server-side) · *CPM* — critical path method · *EVM* — earned value management · *WBS* — work breakdown structure · *RAID* — risks, issues, assumptions, decisions · *Roll-up* — aggregation up the portfolio/program tree · *Integrity layer* — freshness/evidence/doctor trust checks · *Feature Studio* — the 100 ranked AI features · *Room* — a realtime subscription scope.

Document index (this repo)

Doc	Use it for
14 — this manual	benefits, tab outcomes, working design, architecture, business case, cost, setup
01 Deployment	hosting modes, env, upgrades, troubleshooting
02 Admin guide	users, roles, templates, settings, metadata
03 User guide	task-level how-tos per module
04 Architecture	stack, process topology, engines, conventions
05 API reference	route contracts
06 Data model	model-by-model reference
07 RBAC matrix	permission × role matrix
08 Reference data	seed/reference datasets
09 Operations runbook	day-2 operations
10 Security	threat model, controls
11 Integrity & trust	freshness/evidence/doctor design
12 Master platform audit	capability audit evidence
AI-FEATURE-ROADMAP	all 100 AI features specified
AI-BYOK-ARCHITECTURE	AI provider/vault design
FLOW-PLANNING-WORKBOOK-IMPORT	workbook import + date rules
EMAIL-REMINDERS-GUIDE	Graph cron setup
ANALYTICS-ENGINE	analytics design
BUSINESS-PLAN-2026-2027	pricing, benchmarks, TCO (§6–7 source)
COMMAND-CENTER-PRD	full product PRD/ERD

End of manual. Feedback → PMO Administrator; corrections land in the same numbered doc suite.